/* ==
ClickHouse Cloud Lab Script
Topic: Testing Columnar Database with Telecom Recharge Data
Level: Beginner to Intermediate
== */

/* 1. Create Database */

```
9  CREATE DATABASE IF NOT EXISTS telecom_lab;
```

Search results... Elapsed: 82ms

✓ CREATE succeeded

/* 2. Use Database */

```
13  USE telecom_lab;
```

Search results...

✓ USE succeeded

/* 3. Drop old table if already exists */

```
17  DROP TABLE IF EXISTS recharge_transactions;
```

Search results... Elapsed: 0.4ms

✓ DROP succeeded

/* 4. Create Columnar Table using MergeTree Engine */

```
CREATE TABLE recharge_transactions
(
    transaction_id UInt32,
    recharge_date Date,
    msisdn String,
    customer_name String,
    city LowCardinality(String),
    operator LowCardinality(String),
    payment_method LowCardinality(String),
    recharge_amount Decimal(10,2)
)
ENGINE = MergeTree
ORDER BY (recharge_date, city);
```

Search results... Elapsed: 168ms

✓ CREATE succeeded

/* 5. Insert Sample Telecom Recharge Data */

```
INSERT INTO recharge_transactions VALUES
(1,'2026-07-01','959111111','Aung Aung','Yangon','MPT','KBZPay',1000),
(2,'2026-07-01','959222222','Su Su','Yangon','ATOM','WavePay',3000),
(3,'2026-07-01','959333333','Mg Mg','Mandalay','Ooredoo','Cash',5000),
(4,'2026-07-02','959444444','Hla Hla','Yangon','MPT','KBZPay',2500),
(5,'2026-07-02','959555555','Kyaw Kyaw','Bago','ATOM','Cash',1500),
(6,'2026-07-02','959666666','Moe Moe','Yangon','MPT','WavePay',3500),
(7,'2026-07-03','959777777','Ei Ei','Mandalay','ATOM','KBZPay',4500),
(8,'2026-07-03','959888888','Zaw Zaw','Yangon','Ooredoo','Cash',6000),
(9,'2026-07-03','959999999','Cherry','Bago','MPT','WavePay',2000),
(10,'2026-07-04','959101010','Ko Ko','Yangon','ATOM','KBZPay',5500);
```

Search results... Elapsed: 160ms Read: 10 rows (0.70 KB)

✓ INSERT succeeded

/* 6. View All Data */

```
SELECT *
FROM recharge_transactions;
```

Search results... Elapsed: 2ms Read: 10 rows (0.49 KB) Table Chart

#	transacti…	recharge_…	msisdn	customer_…	city	operator	payment_m…	recharge_…
1	3	2026-07-01	959333333	Mg Mg	Mandalay	Ooredoo	Cash	5000
2	1	2026-07-01	959111111	Aung Aung	Yangon	MPT	KBZPay	1000
3	2	2026-07-01	959222222	Su Su	Yangon	ATOM	WavePay	3000
4	5	2026-07-02	959555555	Kyaw Kyaw	Bago	ATOM	Cash	1500
5	4	2026-07-02	959444444	Hla Hla	Yangon	MPT	KBZPay	2500
6	6	2026-07-02	959666666	Moe Moe	Yangon	MPT	WavePay	3500
7	9	2026-07-03	959999999	Cherry	Bago	MPT	WavePay	2000
8	7	2026-07-03	959777777	Ei Ei	Mandalay	ATOM	KBZPay	4500
9	8	2026-07-03	959888888	Zaw Zaw	Yangon	Ooredoo	Cash	6000
1…	10	2026-07-04	959101010	Ko Ko	Yangon	ATOM	KBZPay	5500

10 rows 1 All rows

/* 7. Check Table Structure */

```
DESCRIBE TABLE recharge_transactions;
```

Search results... Elapsed: 0.9ms Read: 8 rows (0.64 KB) Table Chart

#…	name	type	default_type	default_exp…	comment	codec_expre…	ttl_express…
1…	transaction…	UInt32					
2…	recharge_da…	Date					
3…	msisdn	String					
4…	customer_na…	String					
5…	city	LowCardinal…					
6…	operator	LowCardinal…					
7…	payment_met…	LowCardinal…					
8…	recharge_am…	Decimal(10,…					

/* 8. Count Total Rows */

```
60  SELECT COUNT(*) AS total_transactions
61  FROM recharge_transactions;
```

Search results...

#…	total_transactions
1…	10

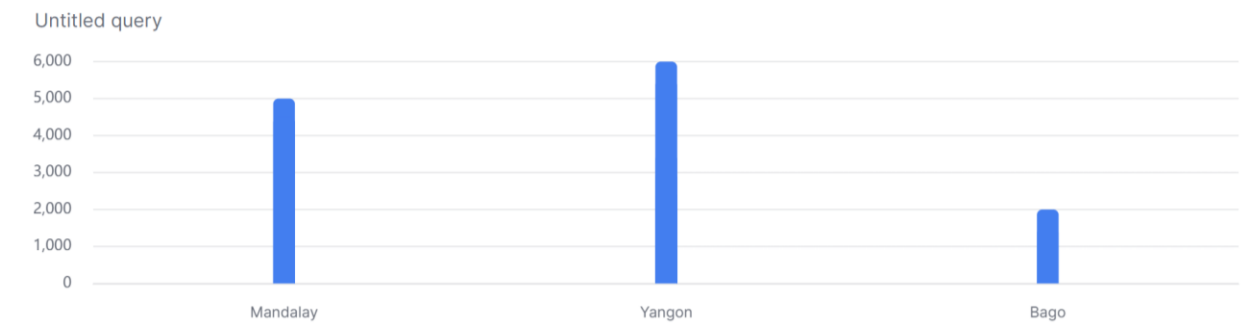
/* 9. Test Columnar Reading
  Only selected columns are read, not the full row.
*/

```
67  SELECT
68      city,
69      recharge_amount
70  FROM recharge_transactions;
```

Search results... Elapsed: 2ms Read: 10 rows (0.14 KB) Table Chart

#	city	recharge_amount
1	Mandalay	5000
2	Yangon	1000
3	Yangon	3000
4	Bago	1500
5	Yangon	2500
6	Yangon	3500
7	Bago	2000
8	Mandalay	4500
9	Yangon	6000
1…	Yangon	5500

Search results... Elapsed: 2ms Read: 10 rows (0.14 KB) Table Chart



/* 10. Filter Data by City */

```
74 SELECT *
75 FROM recharge_transactions
76 WHERE city = 'Yangon';
```

Search results... Elapsed: 3ms Read: 10 rows (0.49 KB) Table Chart

#…	transacti…	recharge_…	msisdn	customer_…	city	operator	payment_m…	recharge_…
1…	1	2026-07-01	959111111	Aung Aung	Yangon	MPT	KBZPay	1000
2…	2	2026-07-01	959222222	Su Su	Yangon	ATOM	WavePay	3000
3…	4	2026-07-02	959444444	Hla Hla	Yangon	MPT	KBZPay	2500
4…	6	2026-07-02	959666666	Moe Moe	Yangon	MPT	WavePay	3500
5…	8	2026-07-03	959888888	Zaw Zaw	Yangon	Ooredoo	Cash	6000
6…	10	2026-07-04	959101010	Ko Ko	Yangon	ATOM	KBZPay	5500

/* 11. Filter Data by Operator */

```
80 SELECT *
81 FROM recharge_transactions
82 WHERE operator = 'MPT';
```

Search results... Elapsed: 4ms Read: 10 rows (0.49 KB) Table Chart

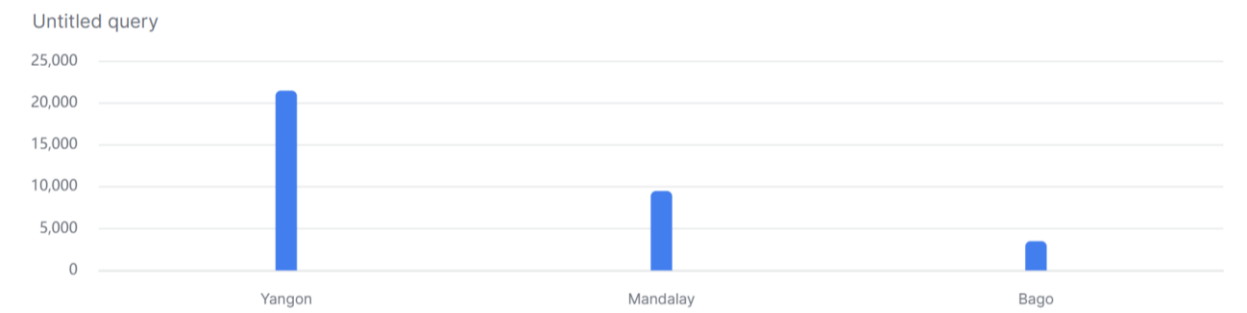
#…	transacti…	recharge_…	msisdn	customer_…	city	operator	payment_m…	recharge_…
1…	1	2026-07-01	959111111	Aung Aung	Yangon	MPT	KBZPay	1000
2…	4	2026-07-02	959444444	Hla Hla	Yangon	MPT	KBZPay	2500
3…	6	2026-07-02	959666666	Moe Moe	Yangon	MPT	WavePay	3500
4…	9	2026-07-03	959999999	Cherry	Bago	MPT	WavePay	2000

/* 12. Total Revenue by City */

```
SELECT
    city,
    SUM(recharge_amount) AS total_revenue
FROM recharge_transactions
GROUP BY city
ORDER BY total_revenue DESC;
```

Search results... Elapsed: 3ms Read: 10 rows (0.14 KB) Table Chart

#…	city	total_revenue
1…	Yangon	21500
2…	Mandalay	9500
3…	Bago	3500

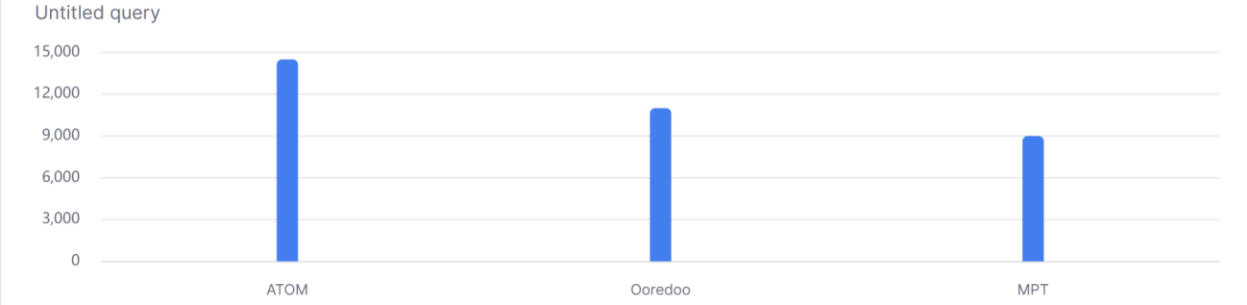


/* 13. Total Revenue by Operator */

```
SELECT
    operator,
    SUM(recharge_amount) AS total_revenue
FROM recharge_transactions
GROUP BY operator
ORDER BY total_revenue DESC;
```

Search results... Elapsed: 4ms Read: 10 rows (0.14 KB) Table Chart

#…	operator	total_revenue
1…	ATOM	14500
2…	Ooredoo	11000
3…	MPT	9000

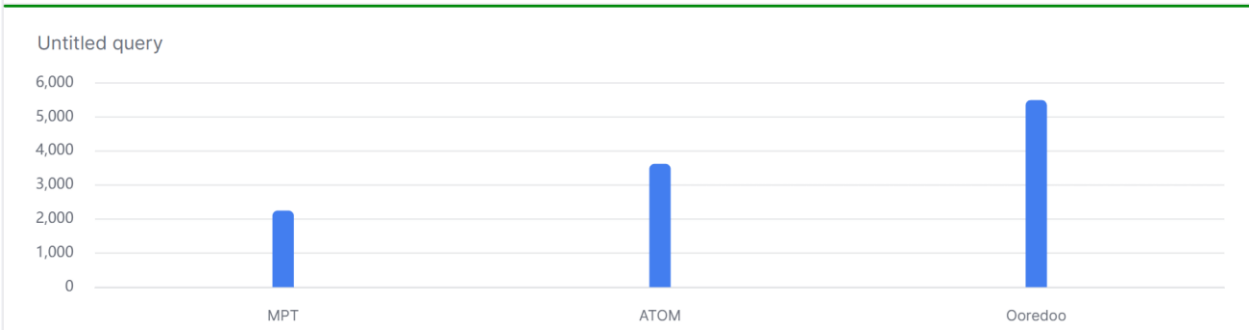


/* 14. Average Recharge by Operator */

```
104 SELECT
105     operator,
106     AVG(recharge_amount) AS average_recharge
107 FROM recharge_transactions
108 GROUP BY operator;
```

Search results... Elapsed: 3ms Read: 10 rows (0.14 KB) Table Chart

#...	operator	average_recharge
1...	MPT	2250
2...	ATOM	3625
3...	Ooredoo	5500

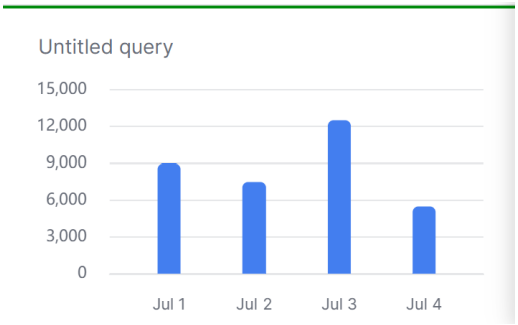


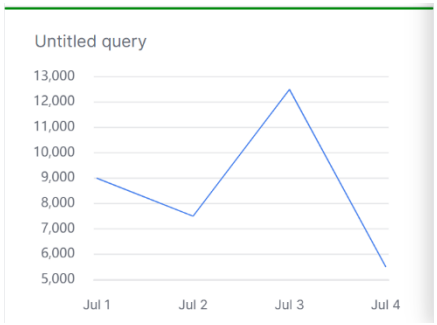
/* 15. Daily Revenue */

```
112 SELECT
113     recharge_date,
114     SUM(recharge_amount) AS daily_revenue
115 FROM recharge_transactions
116 GROUP BY recharge_date
117 ORDER BY recharge_date;
```

Search results... Elapsed: 4ms Read: 10 rows (0.10 KB) Table Chart

#...	recharge_date	daily_revenue
1...	2026-07-01	9000
2...	2026-07-02	7500
3...	2026-07-03	12500
4...	2026-07-04	5500





/* 16. City and Operator Revenue Report */

```sql
SELECT
    city,
    operator,
    COUNT(*) AS total_transactions,
    SUM(recharge_amount) AS total_revenue,
    AVG(recharge_amount) AS average_recharge,
    MAX(recharge_amount) AS highest_recharge,
    MIN(recharge_amount) AS lowest_recharge
FROM recharge_transactions
GROUP BY
    city,
    operator
ORDER BY total_revenue DESC;
```

#…	city	operator	total_tran…	total_reve…	average_re…	highest_re…	lowest_rec…
1…	Yangon	ATOM	2	8500	4250	5500	3000
2…	Yangon	MPT	3	7000	2333.33333…	3500	1000
3…	Yangon	Ooredoo	1	6000	6000	6000	6000
4…	Mandalay	Ooredoo	1	5000	5000	5000	5000
5…	Mandalay	ATOM	1	4500	4500	4500	4500
6…	Bago	MPT	1	2000	2000	2000	2000
7…	Bago	ATOM	1	1500	1500	1500	1500

/* 17. Top 5 Recharge Transactions */

```sql
SELECT *
FROM recharge_transactions
ORDER BY recharge_amount DESC
LIMIT 5;
```

Search results... Elapsed: 3ms Read: 20 rows (0.57 KB) Table Chart

#…	transact…	recharge…	msisdn	customer…	city	operator	payment_…	recharge…
1…	8	2026-07-…	959888888	Zaw Zaw	Yangon	Ooredoo	Cash	6000
2…	10	2026-07-…	959101010	Ko Ko	Yangon	ATOM	KBZPay	5500
3…	3	2026-07-…	959333333	Mg Mg	Mandalay	Ooredoo	Cash	5000
4…	7	2026-07-…	959777777	Ei Ei	Mandalay	ATOM	KBZPay	4500
5…	6	2026-07-…	959666666	Moe Moe	Yangon	MPT	WavePay	3500

/* 18. Top Customers by Recharge Amount */

```sql
SELECT
    customer_name,
    msisdn,
    SUM(recharge_amount) AS total_recharge
FROM recharge_transactions
GROUP BY
    customer_name,
    msisdn
ORDER BY total_recharge DESC;
```

#	customer_name	msisdn	total_recharge
1	Zaw Zaw	959888888	6000
2	Ko Ko	959101010	5500
3	Mg Mg	959333333	5000
4	Ei Ei	959777777	4500
5	Moe Moe	959666666	3500
6	Su Su	959222222	3000
7	Hla Hla	959444444	2500
8	Cherry	959999999	2000
9	Kyaw Kyaw	959555555	1500
1…	Aung Aung	959111111	1000

/* 19. Payment Method Analysis */

```sql
SELECT
    payment_method,
    COUNT(*) AS total_transactions,
    SUM(recharge_amount) AS total_amount
FROM recharge_transactions
GROUP BY payment_method
ORDER BY total_amount DESC;
```

Search results... Elapsed: 4ms Read: 10 rows (0.14 KB) Table Chart

#…	payment_method	total_transactions	total_amount
1…	KBZPay	4	13500
2…	Cash	3	12500
3…	WavePay	3	8500

/* 20. Explain Query Plan */

```
166 EXPLAIN
167 SELECT
168     city,
169     SUM(recharge_amount) AS total_revenue
170 FROM recharge_transactions
171 GROUP BY city;
```

Search results... Elapsed: 2ms Read: 4 rows (0.22 KB) Table Chart

#	explain
1	Expression ((Project names + Projection))
2	Aggregating
3	Expression ((Before GROUP BY + Change column names to column identifiers))
4	ReadFromMergeTree (default.recharge_transactions)

/* 21. Compare Row Read vs Column Read */

/* Reads all columns */

```
177 SELECT *
178 FROM recharge_transactions;
```

Search results... Elapsed: 2ms Read: 10 rows (0.49 KB) Table Chart

#	transact…	recharge…	msisdn	customer…	city	operator	payment_…	recharge…
1	3	2026-07-…	959333333	Mg Mg	Mandalay	Ooredoo	Cash	5000
2	1	2026-07-…	959111111	Aung Aung	Yangon	MPT	KBZPay	1000
3	2	2026-07-…	959222222	Su Su	Yangon	ATOM	WavePay	3000
4	5	2026-07-…	959555555	Kyaw Kyaw	Bago	ATOM	Cash	1500
5	4	2026-07-	959444444	Hla Hla	Yangon	MPT	KBZPay	2500

/* Reads only required columns */

```
181 SELECT
182     city,
183     SUM(recharge_amount) AS total_revenue
184 FROM recharge_transactions
185 GROUP BY city;
```

Search results... Elapsed: 2ms Read: 10 rows (0.14 KB) Table Chart

#	city	total_revenue
1	Mandalay	9500
2	Yangon	21500
3	Bago	3500

/* 22. Duplicate Data for Performance Testing
Run this several times to increase row count.
*/

```
INSERT INTO recharge_transactions
SELECT
    transaction_id + 1000,
    recharge_date,
    msisdn,
    customer_name,
    city,
    operator,
    payment_method,
    recharge_amount
FROM recharge_transactions;
```

Search results... Elapse

✓ INSERT succeeded

/* 23. Check New Row Count */

```
SELECT COUNT(*) AS total_rows
FROM recharge_transactions;
```

Search results...

#...	total_rows
1...	40

/* 24. Performance Test Query */

```
SELECT
    city,
    COUNT(*) AS total_transactions,
    SUM(recharge_amount) AS total_revenue,
    AVG(recharge_amount) AS average_recharge
FROM recharge_transactions
GROUP BY city
ORDER BY total_revenue DESC;
```

Search results... Elapsed: 6ms Read: 40 rows (0.51 KB) Table Chart

#...	city	total_transactions	total_revenue	average_recharge
1...	Yangon	24	86000	3583.3333333333335
2...	Mandalay	8	38000	4750
3...	Bago	8	14000	1750

/* 25. Cleanup Script
  Uncomment when you want to delete the lab objects.
*/

/*
DROP TABLE IF EXISTS recharge_transactions;
DROP DATABASE IF EXISTS telecom_lab;
*/